# Entering data

Eighteen rules that decide whether your analysis takes an afternoon or a fortnight

Worked through on a dental example: two implant systems, one practice

# Why an hour spent here saves a week later

- Most of the time in any analysis goes on cleaning the data, not on statistics.
- Every rule below exists because someone lost a week to breaking it.
- The spreadsheet is not the report. It has exactly one job: to be read by software.
- A clean table can be analysed in any program — R, SPSS, Stata, JMP — without touching it again.

**The one sentence**

A data sheet should be boring: one row per patient, one column per variable, one number per cell, and nothing else in the file.

BEFORE

# Data from hell

| Comparing implant systems A and B | | | | | | | |
|---|---|---|---|---|---|---|---|
| Patient | sex | age | tooth | probing depth | BOP | implant date | complication |
| | | (years) | | (mm) | | | |
| | | | | | | | |
| System A | | | | | | | |
| 1 | Male | 54 | 36 | 4mm | yes | 12.3.2019 | none |
| 2 | w | sixty | tooth 46 | >6 | Y | March 2019 | ? |
| 3 | female | 47 | 3.6 | 5-6 | 1 | 2019 | periimplantitis, bone loss |
| | | | | | | | |
| System B | | | | | | | |
| 1 | M | 61 | 24 | 3 | no | 2020-05-01 | - |
| 2 | Frau | 58 | 14 | n.a. | ja | 1.6.20 | none |
| Mean | | 56 | | 4.5 | | | |

*Every shaded cell is a problem. There are eleven distinct rules broken on this one sheet — and it is not an exaggerated example.*

AFTER

# Data from heaven

| id | system | sex | age | tooth | depth_mm | bop | date_impl | compl |
|---|---|---|---|---|---|---|---|---|
| 1 | 0 | 1 | 54 | 36 | 4.0 | 1 | 2019-03-12 | 0 |
| 2 | 0 | 0 | 60 | 46 | 6.5 | 1 | 2019-03-01 | |
| 3 | 0 | 0 | 47 | 36 | 5.5 | 1 | 2019-06-15 | 1 |
| 4 | 1 | 1 | 61 | 24 | 3.0 | 0 | 2020-05-01 | 0 |
| 5 | 1 | 0 | 58 | 14 | | 1 | 2020-06-01 | 0 |

*Same study, same patients. One row per patient, one variable per column, one value per cell.*

### Nothing here needs explaining to a computer

No units, no words, no symbols, no blank rows, no totals — and every column can be added, averaged or plotted the moment it is read in.

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# Clean the layout

Rules 1 to 5: what a spreadsheet must not contain.

RULE 1

# No title above the table

### Wrong

The first row of the file is the project title, spanning the sheet. The software reads it as the names of the columns.

### Right

The first row is the header row and nothing else. Put the project title in the file name, or in the data dictionary.

### Why it matters

Every program takes row 1 as the variable names. A title there gives you one variable called "Comparing implant systems A and B" and seven called X2 to X8.

RULE 2

# No second header row for units

| Patient | age | probing depth |
|---|---|---|
| | (years) | (mm) |
| 1 | 54 | 4.0 |

### Put the unit in the name instead

depth_mm, age_years, time_min. The unit then travels with the variable into every table and every plot you ever make.

### Why

A row of text under the header turns the whole column into text. Ages become words, and no mean can be taken.

RULE 3

# No blank or separator rows

- No empty row between the two systems.
- No row saying "System B" in the middle of the data.
- No empty row for readability anywhere.

### What to do instead

Add a column. A variable called system with 0 for A and 1 for B tells the software what the blank rows were trying to say — and can then be used in the analysis.

### Why

Many programs stop reading at the first empty row. Others read the blanks as patients with no data, and your n silently grows.

RULE 4

# No summary rows at the bottom

### Wrong

A final row with the mean, the total, or a count — sitting in the same columns as the patients.

### Right

The file ends with the last patient. Summaries are computed by the software, every time, from the raw values.

### Why it matters

A mean row is read as an extra patient. Your n goes from 20 to 21, the mean of the means is wrong, and nobody notices until a reviewer does. It is also the last thing to update when a value changes.

RULE 5

# Give every patient an ID, in the first column

| id | system | sex | age |
|---|---|---|---|
| 1 | 0 | 1 | 54 |
| 2 | 0 | 0 | 60 |
| 3 | 0 | 0 | 47 |

- Unique. Sequential. Never reused.
- Not the name, not the chart number.
- Keeps its meaning after sorting.
- Lets two files be merged later.

### The moment you will be grateful

When a value looks wrong and you have to go back to the paper chart, the ID is the only bridge between the anonymous file and the patient. Keep the ID-to-name list separately, and locked.

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# Name and code the variables

Rules 6 to 9 and 14: what belongs in a cell.

RULE 6

# Valid variable names

### Rules

Short and memorable.

Unique — duplicates are not allowed.

No spaces and no symbols: # @ $ % & are out.

Never start with a number.

Some programs ignore case: DepthMM, depthmm and Depthmm may all be the same name.

| avoid | use |
|---|---|
| `Probing depth (mm)` | `depth_mm` |
| `1st visit` | `visit1` |
| `BOP %` | `bop_pct` |
| `tooth #` | `tooth` |

RULE 7

# Code categories as numbers

| sex (as entered) | sex (coded) |
|---|---|
| Male | 1 |
| w | 0 |
| Frau | 0 |
| m | 1 |
| female | 0 |

*Five spellings of two categories. The computer sees five.*

### Use 0 and 1

For two groups, code 0 and 1 and let 0 be the reference — the untreated arm, the control, the absent finding. Every regression output then reads as "compared with the reference".

RULE 8

# Never mix symbols or units with the value

| as entered | what it should be |
|---|---|
| 4mm | 4.0 |
| > 6 | 6.5 |
| 5-6 | 5.5 |
| 180cm | 180 |
| ca. 45 | 45 |

### One symbol is enough

A single "4mm" anywhere in the column turns all 200 values into text.

### Ranges need a decision

">6" and "5-6" are not measurements. Decide the rule before entry, and write it in the data dictionary.

RULE 9

# One variable per column

| as entered |
| --- |
| Blood pressure |
| 120/80 |
| 140/90 |

→

| sbp | dbp |
| --- | --- |
| 120 | 80 |
| 140 | 90 |
| | |

### The same applies to names and labels

"Control1, Control2, Experiment1, Experiment2" is two variables pretending to be one: an ID and a group. Split them into id and group. And never put a second measurement in the same cell as the first.

RULE 14

# Ordinal categories go in one column

| pain_mild | pain_mod | pain_severe |
|---|---|---|
| 1 | 0 | 0 |
| 0 | 1 | 0 |
| 0 | 0 | 1 |

| pain |
|---|
| 1 |
| 2 |
| 3 |

### Why one column

The three categories are mutually exclusive, so three columns store the same information three times and invite contradictions — a patient marked 1 in two of them. One column cannot contradict itself, and the order 1 < 2 < 3 is preserved for an ordinal analysis.

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# One row per patient

Rules 11 and 12: the shape of the table.

RULE 11

# One patient, one row — never copied

- Do not copy a patient's row into a second block to run a subgroup analysis.
- Do not keep a "men" sheet and a "women" sheet.
- Do not keep one sheet per implant system.

### Instead

Keep one table, add the grouping variable, and let the software select. Subgroups are one line of code; duplicated rows are a permanent source of error.

### Why

A copied patient is counted twice in every total. And when you later correct a value, you will correct it in one copy and not the other — which is worse than the original error, because now the file disagrees with itself.

RULE 12

# Repeated measurements: long or wide

A patient measured at several visits can be stored two ways. Both are legitimate — decide once, and be consistent.

**LONG — one row per patient and visit**

| id | date | depth_mm |
|---|---|---|
| 1 | 2025-01-02 | 4.0 |
| 1 | 2025-07-03 | 3.5 |
| 1 | 2026-01-04 | 3.0 |
| 2 | 2025-03-01 | 5.5 |
| 2 | 2025-09-02 | 5.0 |

**WIDE — one row per patient**

| id | depth1 | depth2 | depth3 |
|---|---|---|---|
| 1 | 4.0 | 3.5 | 3.0 |
| 2 | 5.5 | 5.0 | |

### Which one?

Two or three visits: wide is simpler. More than that, or visits at irregular times: long. R converts between them, so the choice is not permanent.

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# Missing, yes/no, and dates

Rules 10, 13, 15 and 16: the cells that go wrong quietly.

RULE 13

# Yes/no and missing values

| as entered | coded |
|---|---|
| no | 0 |
| y | 1 |
| Yes | 1 |
| N | 0 |
| don't know | (blank) |

### 0 for no, 1 for yes

Always. Never leave a cell empty to mean "no" — an empty cell means "not known", and the two must never be confused.

### Missing = empty cell

Do not type ?, *, -, n.a. or NA. Any of them turns the column into text. Leave it empty and the software reads it as missing.

RULE 15

# Dates

### One format, everywhere

Best is 2019-03-12: year, month, day. It sorts correctly, it is unambiguous between Swiss and American readers, and every program understands it.

### Never mix

12.3.2019, March 2019, 2019, 1.6.20 and "last year" in one column is five formats and no date at all.

- Format the column as a date in Excel before entry (Format, Cells, Date) — so a typo is visible immediately.
- Compute ages and intervals from the dates later. Do not enter both a date and an age: they will disagree.
- Two-digit years are how 1.6.20 becomes 2020 in one program and 1920 in another.

RULE 10

# No graphs and no statistics in the data file

- No chart floating over the cells.
- No mean, SD or count in a corner.
- No coloured cells carrying meaning.
- No comments in the margin.

### Colour is not data

If a yellow cell means "re-measure this one", the meaning is lost the moment the file is exported. Add a column instead.

### Keep two files

One file holds the raw data and nothing else. A second file — or better, a script — holds the analysis. Then the raw file never changes, and every figure can be reproduced from it.

RULE 16

# Several files are fine — if the ID matches

| id | system | sex | age |
|---|---|---|---|
| 1 | 0 | 1 | 54 |
| 2 | 0 | 0 | 60 |
| 3 | 0 | 0 | 47 |

+

| id | depth_mm | bop |
|---|---|---|
| 1 | 4.0 | 1 |
| 2 | 6.5 | 1 |
| 3 | 5.5 | 1 |

### The ID is the join

Clinical data can be entered by one person and radiographic measurements by another, in separate files, at separate times. As long as both carry the same id column, they can be merged in one command. Without it, the merge is done by hand — and done wrong.

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# Protect it and document it

Rules 17 and 18: what happens after entry.

RULE 17

# Confidentiality

- Store the file in a locked, access-controlled place. Back it up daily or weekly.
- Before sending data to a statistician, remove: names, initials, insurance and AHV numbers, chart numbers, addresses.
- Remove actual dates too — dates of birth, of surgery, of admission.
- Send an age or an interval in days instead of the dates themselves.

**Keep one key, and keep it separate**

The link between id and patient stays in the practice, in a separate file, protected.

RULE 18

# Write a data dictionary

| variable | meaning |
|---|---|
| id | Sequential case number |
| system | Implant system: 0 = A, 1 = B |
| sex | 0 = female, 1 = male |
| age | Age in years at implant placement |
| tooth | FDI tooth number |
| depth_mm | Probing depth in mm, deepest site, 12 months |
| bop | Bleeding on probing: 0 = no, 1 = yes |
| compl | Complication within 12 months: 0 = no, 1 = yes |

### Write it while you enter, not afterwards

In two years you will not remember whether 0 meant system A or system B — and neither will the person who has to finish your paper.

# And then it is one line to read in

```
implants <- read.csv("implants.csv")    # save the sheet as CSV first

str(implants)                           # what did R actually get?
summary(implants$depth_mm)
table(implants$system, implants$bop)
```

### str() is your proof

If depth_mm comes back as num, the column was clean. If it comes back as chr, there is a "4mm" or a "?" hiding somewhere in it. Check every column on the day you finish entering, not on the day you start analysing.